

Atharva Katurde

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PROFILE

Final-year MSc Artificial Intelligence student at MTU Cork (First Class Honours) with a genuine passion for the human side of technology. I am excited by roles where I can grow at the intersection of AI and business, helping organisations understand what AI can actually do for them, not just what is technically possible. My background spans published IEEE and Springer research, production-deployed AI systems, and real client-facing experience from my internship at Ai India Innovations. I am a fast learner who thrives on taking complex ideas and making them understandable and useful for the people who rely on them. Currently seeking graduate roles in AI consulting, solutions, or technology advisory in Ireland.

EDUCATION

MSc Artificial Intelligence

Sept 2025 to June 2026

Munster Technological University · Cork, Ireland

- Result: In progress | First Class Honours (1:1) achieved in Semester 1
- Thesis: *Generative AI for Resilient Decision Support: A Hierarchical Policy Distillation Framework*
- Core modules: NLP, Practical Machine Learning, Big Data Processing, Knowledge Representation, Agile/Scrum

BEng Computer Engineering (Hons: Data Science)

Sept 2022 to May 2025

Savitribai Phule Pune University · India

- First Class Honours | CGPA 8.86/10
- Team Lead, Smart India Hackathon 2024 | National Finalist (directed end-to-end AI product development)
- Student Lead, NSS | coordinated 60 volunteers across community outreach programmes
- Selected for government-funded IDE Bootcamp at IIM Sambalpur focused on design thinking and startup scaling

EXPERIENCE

Software Engineer | Internship

March 2025 to July 2025

Ai India Innovations · Pune, India · On-site

- Analysed client use cases to select optimal LLM architectures, translating complex AI research into actionable business solutions
- Shipped a Python document-intelligence module (OCR, PyPDF, LLM inference) processing 100+ PDFs/day | delivered 2 weeks ahead of schedule
- Architected a RAG chatbot end-to-end: vector database schema, embedding pipeline, and semantic retrieval layer for context-aware responses
- Deployed hybrid LLM inference (local Llama for privacy, Mistral API for throughput) across a 3-engineer cross-functional team
- Developed technical pitch decks and executive meeting highlights to support leadership during client proposals

Customer Service Team Member

Jan 2026 to Present

McDonald's · Cork, Ireland

- High-pressure customer service and team operations in a fast-paced environment | building communication and adaptability skills

PROJECTS

Digital Twin Portfolio Agent

React · JavaScript · Gemini API · RAG · Vercel

- Built and shipped a live agentic portfolio product: React/JS frontend with a Gemini-powered AI agent grounded in a structured personal knowledge base, handling real recruiter queries in production
- Engineered prompt architecture, tool-call logic, and edge-case handling to maintain agent coherence in a zero-tolerance public environment

Global Market Intelligence & Risk Analysis Suite

Python · n8n · Llama 3 · PostgreSQL · Docker · Power BI

- Problem:** Compliance analysts spending 10+ hours/week manually auditing financial news before strategic decisions
- Solution:** Multi-agent n8n pipeline with parallel streams | a loop agent (Llama 3) scoring each article for sentiment and risk, and a macro agent generating real-time Market Pulse; results stored in Medallion-schema PostgreSQL for full auditability
- Outcome:** Power BI Traffic Light dashboard enabling same-day compliance decisions, replacing a manual multi-step process entirely

BoloTech | Assistive Communication App

Flutter · OpenAI Whisper · WebRTC · Firebase · Python

- Conducted empathy interviews with 15 users before writing a single line of code | built real-time STT/TTS (OpenAI Whisper, 93%+ accuracy) for English, Hindi, and Marathi
- Led a 4-person team through full development lifecycle; low-latency WebRTC pipeline and Firebase auth; MVP secured industry sponsorship

- Multi-agent system (Planner, Writer, Editor, Visual agents) generating 2,500+ word SEO-optimised articles with automatically generated visuals
- Deployed to production with Supabase auth, role-based approval workflows, and multi-format export (PDF, DOCX, Markdown)

RESEARCH & PUBLICATIONS

Enhancing Neural Architecture Search: A Comparative Optimization Framework	Springer (ICCI 2024) · Aug 2025
Surrogate modelling and RL-based strategies improving NAS computational efficiency on ImageNet, CIFAR-10, LibriSpeech	
Enhanced Classification of Astronomical Images Using Optimised Neural Networks via NAS	IEEE · Jun 2025
7-10% accuracy improvement and 30% reduction in inference time for large-scale astronomical image classification	
Enhancing Recommendations with Adaptive Multi-Modal Generative Models	IEEE · Jun 2025
AMGR framework combining GANs, VAEs, and transformers for recommendation system accuracy	
SecureSense: AI/ML Based Anomaly Detection Tool	IEEE · Jul 2024
Behaviour-Based Anomaly Detection Systems (BBADS) for proactive cybersecurity defence	
Data Compression for Backbone Networks	Journal of Multimedia Technology · May 2024
Optimised network throughput and energy consumption via advanced lossless and lossy compression algorithms	

SKILLS

Languages & Frameworks	Python (Advanced), JavaScript, React, SQL, REST APIs, FastAPI
AI & Agentic Systems	LLMs, Prompt Engineering, RAG, Multi-Agent Systems (CrewAI), NLP, PyTorch, Agentic Pipelines
AI-Native Workflow	Cursor, Claude, GitHub Copilot, Gemini API, n8n, Lovable
Infrastructure	Docker, Git, PostgreSQL, Supabase, Firebase, Microsoft Azure, Vercel
Business & Consulting	Technical Translation, Stakeholder Communication, Use-Case Analysis, Requirements Gathering, Agile/Scrum
Practices	Responsible AI, GDPR Awareness, Technical Documentation, OOP, SDLC

ACHIEVEMENTS & CERTIFICATIONS

- **Microsoft Certified: AI Business Professional (AB-730)** · Apr 2026 | Business-facing AI certification covering responsible AI, generative AI, and real-world business applications
- **Oracle Certified Generative AI Engineer** | Professional certification in generative AI system design and deployment
- **Microsoft Azure Fundamentals (AZ-900 Track)** · Mar 2026 | Cloud fundamentals covering Azure compute, storage, networking, and AI services
- **Best Paper Award | IEEE 2025** · Enhanced Classification of Astronomical Images Using Optimised Neural Networks via NAS
- **Smart India Hackathon 2024 | National Finalist** · Directed technical team of 6 to national finals; full AI product development lifecycle from prototype to presentation
- **Best Technical Presenter | Technothon 2024** · Recognised for clarity and impact in technical communication
- **Hackathon Winner 2024** (1st place, 20 teams) · AI-powered solution judged best in show by industry panel
- **NSS Chairperson 2024** · Led 60 volunteers across social welfare and community outreach programmes
- **IIM Sambalpur IDE Bootcamp** · Government-funded intensive on design thinking, startup scaling, and product-market fit
- **1,500+ LinkedIn Followers** · Active thought leader publishing on AI adoption, agentic systems, and responsible AI